Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Science, Technology, Engineering and Mathematics

School of Mathematics

JS and SS Mathematics **Semester 2 2022**
SS Theoretical Physics

MAU44400 — Quantum Field Theory

Wednesday 4 May RDS Main Hall 09:30 - 12:30

Professor Samson L. Shatashvili

Instructions to candidates:

Credit will be given for the best two questions from Part A (Questions 1, 2 and 3) and the best two questions from Part B (Questions 4, 5 and 6).

All questions have equal weight.

Additional instructions for this examination:

Formulae and Tables tables are available from the invigilators, if required. Non-programmable calculators are permitted for this examination,—please indicate the make and model of your calculator on each answer book used.

You may not start this examination until you are instructed to do so by the Invigilator.

Part A

1. Scalar QED: 25 marks

Consider the EM fields coupled to charged matter, namely a complex scalar field $\Phi(x)$ of electric charge e. Altogether, with electromagnetic field Lagrangian is:

$$\mathcal{L} = \mathcal{L}_{EM} + \mathcal{L}_{mat} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + (D^\mu\Phi)^* D_\mu\Phi - m^2\Phi^*\Phi$$

$$\mathcal{L}_{EM} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu}$$

$$\mathcal{L}_{mat} = (D^\mu\Phi)^* D_\mu\Phi - m^2\Phi^*\Phi$$

where

$$D_\mu\Phi = (\partial_\mu + ieA_\mu)\Phi, \qquad (D_\mu\Phi)^* = (\partial_\mu - ieA_\mu)\Phi^*$$

are covariant derivatives. Gauge transformations are:

$$\Phi \to e^{i\alpha}\Phi, \qquad \Phi^* \to e^{-i\alpha}\Phi^*, \qquad A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha$$

a) 5 marks: Write down the equation of motion for all fields in a covariant from (using covariant derivatives).

b) 5 marks: Write down the electric current which by definition is:

$$J^\mu = -\frac{\partial\mathcal{L}}{\partial A_\mu}$$

in a manifestly gauge-invariant form and verify its conservation:

$$\partial_\mu J^\mu = 0$$

(as long as the scalar fields satisfy their equations of motion).

c) 5 marks: Show that the Noether stress-energy tensor for the matter sector, for Φ fields, is:

$$T^{\mu\nu}_{mat} = (D^\mu\Phi)^* D^\nu\Phi + (D^\nu\Phi)^* D^\mu\Phi - \eta^{\mu\nu}((D_\mu\Phi)^* D^\mu\Phi - m^2\Phi^*\Phi)$$

d) 10 marks: Use the scalar fields equations of motion and the non-commutativity of covariant derivatives

$$[D_\mu, D_\nu]\Phi = ieF_{\mu\nu}\Phi$$

to show that

$$\partial_\mu T^{\mu\nu}_{mat} = F^{\nu\lambda}J_\lambda$$

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2. γ matrices

Assume the anti-commutation relations

$$\gamma^\mu \gamma^\nu \; + \; \gamma^\nu \gamma^\mu \; = \; 2\eta^{\mu\nu}$$

and the definition

$$S^{\mu\nu} \; = \; -S^{\nu\mu} \; = \frac{1}{4i}[\gamma^\mu, \gamma^\nu]$$

but do not assume the specific form of the 4×4 matrices γ^μ and $S^{\mu\nu}$.

(a) 10 marks: Show that:

$$\left[S^{\kappa\lambda}, \gamma^\mu\right] \; = \; -i\eta^{\lambda\mu}\gamma^\kappa \; + \; i\eta^{\kappa\mu}\gamma^\lambda$$

(b) 10 marks: Further, show that:

$$\left[S^{\kappa\lambda}, S^{\mu\nu}\right] \; = \; i\eta^{\lambda\nu}S^{\kappa\mu} \; - \; i\eta^{\lambda\mu}S^{\kappa\nu} \; + \; i\eta^{\kappa\mu}S^{\lambda\nu} \; - \; i\eta^{\kappa\nu}S^{\lambda\mu}$$

(c) 5 marks: Consider the matrix $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$:

Show that γ^5 anticommutes with each of the γ^μ matrices, $\gamma^5\gamma^\mu = -\gamma^\mu\gamma^5$ and that $(\gamma^5)^2 = 1$.

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3. Current-current commutators in 2 space-time dimensions: 25 marks

Consider the left-handed fermions in 2 space-time dimensions (t, x): $\psi_L = \frac{1}{2}(1-\gamma_5)\psi_D$:

Define the current:

$$J_0^\epsilon(t,x) = \bar{\psi}_L(t, x+\epsilon)\gamma^0\psi_L(t, x-\epsilon) = \psi_L^+(x+\epsilon)\psi_L(x-\epsilon)$$

(a) **15 marks:** Use the canonical equal time (anti)commutation relations for fermions $\{\psi_L(x), \psi_L^+(y)\} = \delta(x-y)$ and compute the equal time commutators for these current:

$$[J_0^\epsilon(t,x), J_0^\epsilon(t,y)]$$

(b) **10 marks:** For the massless case:

$$\langle 0|\psi_L^+(t,x)\psi_L(t,y)|0\rangle = \frac{1}{x-y}$$

Evaluate

$$\langle 0|[J_0^\epsilon(t,x), J_0^\epsilon(t,y)]|0\rangle$$

and its limit for $\epsilon \to 0$.

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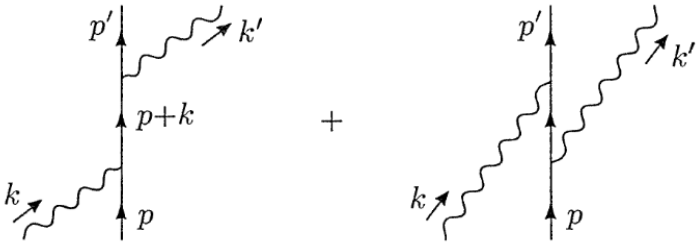
Part B

4. QED at tree level - Compton scattering: 25 marks

a) 15 marks: Write the expression for Compton scattering amplitude $i\mathcal{M}$ (two tree level diagrams for $e^-\gamma \to e^-\gamma$).



b) 10 marks: Replace one of the photon polarisation vectors, f.e. $\epsilon^\mu(k')$, by k'^μ and show that the result for sum of these two diagrams vanishes, thus prove the Ward identity $k'_\mu \mathcal{M}^\mu = 0$ for Compton scattering.

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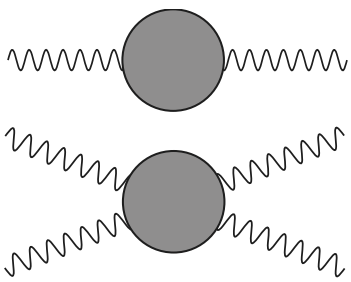
5. Scalar QED: 25 marks

Consider scalar QED in four dimensions. Scalar QED is an interacting theory of photon with complex scalar Φ:

$$\mathcal{L} = \mathcal{L}_{EM} + \mathcal{L}_{mat} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + (D^{\mu}\Phi)^{*}D_{\mu}\Phi - m^{2}\Phi^{*}\Phi$$

Here $D_\mu\Phi = (\partial_\mu + ieA_\mu)\Phi$. Scalar QED is gauge invariant, like QED, under $\Phi \to e^{i\alpha}\Phi, \Phi^* \to e^{-i\alpha}\Phi^*, A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha$.

Feynman diagrams for scalar QED are similar to QED except the three-vertex is $-ie(p+p')^\mu$ instead of $-ie\gamma^\mu$ (where p and p' are momenta of scalar external lines), four-vertex is $2ie^2\eta^{\mu\nu}$, and scalar propagator $\frac{i}{p^2-M^2+i\epsilon}$ replaces the fermion propagator.



a) 15 marks: What is the superficial divergence of each diagram above?

b) 10 marks: Use the Ward Identity $k^\mu\Pi_{\mu,\ldots}(k,\ldots) = 0$ to find out the actual divergence type of each diagram $[\Pi_{\mu,\ldots}(k,\ldots)]$.

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6. Loop integrals: 25 marks

a) 5 marks: Show that for $m > 2$:

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{[l_E^2 + \Delta]^m} = \frac{1}{(4\pi)^2} \frac{1}{(m-1)(m-2)} \frac{1}{\Delta^{m-2}}$$

b) 5 marks Show that for $m > 3$:

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{l^2}{[l_E^2 + \Delta]^m} = \frac{1}{(4\pi)^2} \frac{1}{(m-1)(m-2)(m-3)} \frac{1}{\Delta^{m-3}}$$

c) 10 marks: For $n \geq 2$ calculate:

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{[l_E^2 + \Delta]^n} =$$

d) 5 marks: For $n \geq 2$ calculate:

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{l^2}{[l_E^2 + \Delta]^n} =$$

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